

Phase 7

Project Documentation Phase

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Phase	Covers
Brainstorming & Ideation	Manual workout tracking problem & proposed AI fitness solution
Requirement Analysis	Software & hardware requirements
Requirement Analysis	Layered backend architecture & overall system components
Project Planning	Module definitions, ER diagram, user flow & MVC pattern
Project Development	Project setup, MongoDB connection & Mongoose schemas
Project Testing	Authentication, Workout & AI API testing and validation
Project Documentation	Code structure, database schemas & API specifications
Project Demonstration	Server execution & live API request/response verification

- **Problem & Requirements** – Explains the manual workout tracking problem, the proposed FitTrack AI backend solution integrated with Google Gemini AI, and the required hardware and software specifications.
- **Design & Architecture** – Describes the 7-layer backend architecture, ER diagram and database structure, user flow, MVC pattern, authenticated user role, and the one-to-many User-to-Workout relationship.
- **Development & Implementation** – Covers project initialization, environment configuration, Node.js and Express.js backend setup, MongoDB database connection, Mongoose validation schemas (**User** and **Workout**), JWT authentication middleware, and Google Gemini AI integration.
- **Testing & Demonstration** – Covers starting the backend server and testing user registration, login, profile retrieval, workout CRUD operations, and AI-generated workout recommendations and fitness insights using Postman and Thunder Client.